

■ Flat

Flat is an attribute that can be assigned to a symbol f to indicate that all expressions involving nested functions f should be flattened out. This property is accounted for in pattern matching.

Flat corresponds to the mathematical property of associativity. ■ For a symbol f with attribute Flat, $f[f[a, b], f[c]]$ is automatically reduced to $f[a, b, c]$. ■ Functions like Plus, Times and Dot are Flat. ■ For a Flat function f , the variables x and y in the pattern $f[x_, y_]$ can correspond to any sequence of arguments. ■ The Flat attribute must be assigned before defining any values for an Flat function. ■ See page 214. ■ See also: Orderless, OneIdentity.